

Rail-Admissible Computation and Treewidth-Based Separation for SAT

Technical Supplement with Full Proofs

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1. Detailed Model Definitions

1.1 Graph-Theoretic Preliminaries

Tree decomposition: A tree decomposition of graph $G = (V, E)$ is a pair $(T, \{B_t\}_{t \in V(T)})$ where T is a tree and $B_t \subseteq V$ are bags such that:

- Every vertex $v \in V$ appears in some bag
- Every edge $\{u, v\} \in E$ is contained in some bag
- For each $v \in V$, the set $\{t : v \in B_t\}$ induces a connected subtree of T

Treewidth: $\text{tw}(G) = \min_{\text{decompositions}} \max_t |B_t| - 1$

Clique-sum: For graphs G_1, G_2 with common clique S , the clique-sum $G_1 \oplus_S G_2$ identifies the vertices of S in both graphs. Standard result: $\text{tw}(G_1 \oplus_S G_2) \leq \max(\text{tw}(G_1), \text{tw}(G_2), |S|-1)$.

1.2 Rail-Admissible Computation Model

Primitive gate: A finite undirected graph G_{prim} with:

- Maximum degree ≤ 3 (the " $d \leq 3$ " local constraint)

- Fixed finite size (constant number of vertices)
- Represents one computational operation

Let $\mathcal{G}_{(\leq 3)}$ denote the class of all such graphs. Any $G \in \mathcal{G}_{(\leq 3)}$ has constant treewidth (bounded-degree + finite size).

Module: A finite undirected graph M obtained by:

- Taking disjoint union of primitive gates
- Adding local wiring edges (preserving degree bounds)
- Designating boundary vertices $\partial M \subseteq V(M)$

Formally: $M = (V, E, \partial M)$ where ∂M is the interface.

Rail-join (thickness- k): Given modules $M_1 = (V_1, E_1, \partial M_1)$ and $M_2 = (V_2, E_2, \partial M_2)$ and separator $S \subseteq \partial M_1 \cap \partial M_2$ with $|S| \leq k$:

$$M_1 \oplus_S M_2 = (V_1 \cup V_2, E_1 \cup E_2 \cup E_S, (\partial M_1 \cup \partial M_2) \setminus S)$$

where:

- Vertices in S are identified between M_1 and M_2
- $E_S = \{\{u, v\} : u, v \in S, u \neq v\}$ (S becomes a clique, "moralization")

k-rail-admissible composition: A family of modules $\{M_i\}_{i \in I}$ with:

1. A tree $T = (I, E(T))$ on the module indices
2. For each edge $(i, j) \in E(T)$, a separator $S_{ij} \subseteq \partial M_i \cap \partial M_j$ with $|S_{ij}| \leq k$
3. Running intersection property: for each vertex v appearing in multiple modules, the set $\{i : v \in V(M_i)\}$ forms a connected subtree of T
4. The global graph G is built by iterative rail-joins along edges of T

Definition (Rail-Admissible): A computation is **k-rail-admissible** if it can be expressed as a k-rail-admissible composition where each primitive gate has degree ≤ 3 .

For concreteness, we focus on $k=12$ (the "thickness ≤ 12 " constraint from the Grammaton framework).

2. Lemma 1: Full Proof

Lemma 1 (Rail Width $\Rightarrow$ Bounded Treewidth).

Let G be the computation graph of a k-rail-admissible composition built from primitive gates with local degree ≤ 3 . Let $w_0 = \max\{\text{tw}(M_i)\}$ be the maximum treewidth of any single module. Then:

$$\text{tw}(G) \leq \max(w_0, k-1)$$

Proof

Setup: We have modules $\{M_i\}_{i \in I}$ arranged in tree $T = (I, E(T))$ with separators $\{S_{ij}\}_{(i,j) \in E(T)}$ where $|S_{ij}| \leq k$ for all edges.

Strategy: Induction on the number of modules $|I| = n$.

Base case ($n=1$): $G = M_1$ is a single module. By definition, $\text{tw}(M_1) \leq w_0$. Since $w_0 \leq \max(w_0, k-1)$, the claim holds.

Inductive hypothesis: Assume for any rail-admissible composition of n modules, $\text{tw} \leq \max(w_0, k-1)$.

Inductive step ($n \rightarrow n+1$): Consider composition with $n+1$ modules. Since T is a tree, there exists a leaf module M_{leaf} connected to exactly one other module. Let:

- R = rest of composition (n modules)
- M_{leaf} = the leaf module
- S = separator between R and M_{leaf} , with $|S| \leq k$

By inductive hypothesis: $\text{tw}(R) \leq \max(w_0, k-1)$

By definition: $\text{tw}(M_{\text{leaf}}) \leq w_0$

The new graph is: $G = R \oplus_S M_{\text{leaf}}$

Apply clique-sum bound:

The standard theorem for treewidth under clique-sum states:

$$\text{tw}(H_1 \oplus_S H_2) \leq \max(\text{tw}(H_1), \text{tw}(H_2), |S| - 1)$$

Proof of clique-sum bound (for completeness): Take optimal tree decompositions $(T_1, \{B^1_i\})$ of H_1 and $(T_2, \{B^2_i\})$ of H_2 . If S is not already a bag in T_1 , add it (width unchanged since $S \subseteq$ some bag by decomposition properties, or we can augment). Similarly for T_2 . Connect these two decompositions at their S -bags. The result is a valid tree decomposition of $H_1 \oplus_S H_2$ with width $\max(\text{tw}(H_1), \text{tw}(H_2), |S| - 1)$.

Applying to our case:

$$\begin{aligned} \text{tw}(G) &= \text{tw}(R \oplus_S M_{\text{leaf}}) \\ &\leq \max(\text{tw}(R), \text{tw}(M_{\text{leaf}}), |S| - 1) \\ &\leq \max(\max(w_0, k-1), w_0, k-1) \\ &= \max(w_0, k-1) \end{aligned}$$

By induction, the bound holds for all n . ■

Corollary: For $k=12$ and constant-size primitive gates, $\text{tw}(G) = O(1)$.

Remark on necessity of tree structure: If modules formed cycles with overlapping interfaces, we could simulate grid-like structures with unbounded treewidth despite local constraints. The tree-of-joins assumption is essential.

3. Lemma 2: Full Proof

Lemma 2 (SAT Hardness via Expanders).

There exists a polynomial-time constructible family $\{F_n\}$ of unsatisfiable 3-CNF formulas on $\Theta(n)$ variables such that:

1. $w_{res}(F_n) \geq cn$ for constant $c > 0$
2. $tw(G_{pr}(F_n)) \geq cn - 1$

where $G_{pr}(F)$ is the primal (variable-interaction) graph of F .

3.1 Construction: Tseitin Formulas on Expanders

Step 1: Expander graph

Let $H_n = (V, E)$ be a d -regular graph on n vertices with:

- Degree $d = 3$ (constant, can be any $d \geq 3$)
- Edge expansion: $h(H_n) \geq h_0 > 0$ (constant)
- Girth $g(H_n) \geq g_0 > 0$ (constant)

Existence: Ramanujan graphs provide explicit polynomial-time constructions meeting these requirements (Lubotzky-Phillips-Sarnak 1988, Margulis 1988).

Step 2: Charge assignment

Define charge function $\chi : V \rightarrow \{0,1\}$ with:

- $\chi(v^*) = 1$ for one designated vertex $v^* \in V$
- $\chi(v) = 0$ for all $v \neq v^*$

Total charge: $\sum_v \chi(v) = 1$ (odd)

Step 3: Tseitin encoding

For each edge $e = \{u, v\} \in E$, introduce Boolean variable x_e .

For each vertex $v \in V$ with incident edges e_1, e_2, e_3 (since $d=3$), encode the parity constraint:

$$x_{e_1} \oplus x_{e_2} \oplus x_{e_3} = \chi(v)$$

Step 4: CNF conversion

Standard gadget converts each XOR constraint to clauses. For $a \oplus b \oplus c = 0$:

$$(\neg a \vee \neg b \vee \neg c), (a \vee b \vee \neg c), (a \vee \neg b \vee c), (\neg a \vee b \vee c)$$

For $a \oplus b \oplus c = 1$, negate appropriately.

Let F_n be the resulting 3-CNF formula.

Properties:

- $|\text{Variables}| = |E| = (3n/2)$ (since $d=3$)
- $|\text{Clauses}| = 4|V| = 4n$
- All clauses have size ≤ 3

3.2 Unsatisfiability

Claim: F_n is unsatisfiable.

Proof: Suppose assignment σ satisfies all clauses. Then for each vertex v , the parity constraint holds:

$$\bigoplus_{e \ni v} \sigma(x_e) = \chi(v).$$

Sum over all vertices:

$$\sum_v [\bigoplus_{e \ni v} \sigma(x_e)] = \sum_v \chi(v) = 1 \pmod{2}$$

But the left side counts each edge exactly twice (once for each endpoint), so:

$$\sum_v [\bigoplus_{e \ni v} \sigma(x_e)] = 2 \cdot (\sum_e \sigma(x_e)) = 0 \pmod{2}$$

Contradiction. Therefore F_n is unsatisfiable. ■

3.3 Resolution Width Lower Bound

Theorem (Ben-Sasson & Wigderson 2001): For Tseitin formulas on expander graphs with constant degree d and expansion $h_0 > 0$:

$$w_{\text{res}}(F_n) \geq c \cdot h_0 \cdot n / d$$

for absolute constant $c > 0$.

For our construction: $d=3$, h_0 constant, therefore $w_{\text{res}}(F_n) \geq c'n$ for $c' = c \cdot h_0 / 3 > 0$.

Intuition: Any resolution refutation must derive clauses that cut through the expander. Edge expansion forces any balanced cut to have size $\Omega(n)$, which translates to clause width $\Omega(n)$ via parity constraints.

3.4 Treewidth Lower Bound

Theorem (Folklore, follows from resolution-width connection): For any CNF formula F :

$$\text{tw}(G_{\text{pr}}(F)) \geq w_{\text{res}}(F) - 1$$

Proof sketch: A tree decomposition of width k provides an elimination order where intermediate formulas during variable elimination have clause width $\leq k+1$. This simulates resolution of width $\leq k+1$. Therefore $w_{\text{res}}(F) \leq \text{tw}(G_{\text{pr}}(F)) + 1$.

For our family:

$$\text{tw}(G_{\text{pr}}(F_n)) \geq w_{\text{res}}(F_n) - 1 \geq c' n - 1 = \Omega(n)$$

Alternative direct argument: The primal graph $G_{\text{pr}}(F_n)$ closely mirrors the edge structure of H_n (variables = edges, constraints couple adjacent edges). Expander graphs have treewidth $\Theta(n)$ (standard result: any balanced separator in an expander has size $\Omega(n)$). Therefore $\text{tw}(G_{\text{pr}}(F_n)) = \Omega(n)$.

This completes Lemma 2. ■

4. Lemma 3: Full Proof

Lemma 3 (Communication Lower Bound).

For the lifted family $\phi_{\{G,q\}}$ on $\Theta(nq)$ variables derived from expander G and strip parameter q , any width- k tree decomposition has an edge (u,v) whose separator S_{uv} requires at least:

$$\Omega(nq / (k+1))$$

bits to correctly evaluate SAT via dynamic programming.

4.1 Lifted Construction

Graph: Take $G \square P_q$ (Cartesian product of expander G with path P_q of length q)

- Vertices: $V(G) \times \{1, \dots, q\}$
- Edges: $\{(v,i), (v,i+1)\}$ for $v \in V(G), i < q$ (path edges) $\{(u,i), (v,i)\}$ for $\{u,v\} \in E(G)$, all i (expander edges at each layer)

Formula $\phi_{\{G,q\}}$: Apply Tseitin construction to each layer i with independent charge assignments (all odd to maintain unsatisfiability). Total: $\Theta(nq)$ variables, $\Theta(nq)$ clauses.

Key property: The formula encodes $\Theta(nq)$ independent parity constraints that must be reconciled across any balanced separator.

4.2 Dynamic Programming Message Model

Fix tree decomposition $(T, \{B_t\})$ of $G_{pr}(\phi_{\{G,q\}})$ with width k .

Message at edge (u,v) : Let $S_{uv} = B_u \cap B_v$ be the separator ($|S_{uv}| \leq k+1$).

The subtree below u computes, for each partial assignment $\alpha : S_{uv} \rightarrow \{0,1\}$, whether its clauses are satisfiable given α . This is encoded as message $M_{\{u \rightarrow v\}} : \{0,1\}^{|S_{uv}|} \rightarrow \{0,1\}$.

Communication cost: The message requires $2^{|S_{uv}|}$ bits in worst case (one bit per possible assignment).

However, we can do better: we need to count **independent linear constraints** over $\mathbb{F}_2$ that cannot be determined from S_{uv} alone.

4.3 Information-Theoretic Lower Bound

Claim: For $\phi_{\{G,q\}}$, there exists an edge (u,v) where $\Theta(nq)$ independent parity constraints cross the separator.

Proof:

Step 1: Balanced separator

By standard centroid decomposition, T has an edge (u,v) that roughly balances the decomposition: $|V_1| \approx |V_2| \approx nq/2$ where V_1, V_2 are the variable sets in the two subtrees.

Step 2: Edge-disjoint constraints

The product structure $G \square P_q$ creates q "layers," each an expander. Any balanced vertex partition induces a cut in each layer. By expansion, each layer's cut has size $\geq h_0 \cdot (n/2) = \Omega(n)$ edges. Summing over q layers: $\Omega(nq)$ crossing edges.

Step 3: Independent parities

Each crossing edge e contributes a parity constraint involving x_e . By girth assumption and product structure, these constraints are (nearly) independent over $\mathbb{F}_2$: learning the parity values on V_1 does not determine parities on V_2 without communicating the crossing edge values.

Formally: The rank (over $\mathbb{F}_2$) of the constraint matrix restricted to crossing edges is $\Omega(nq)$.

Step 4: Separator capacity

The separator S_{uv} has $|S_{uv}| \leq k+1$ variables. Any assignment to these variables fixes at most $k+1$ linear constraints over $\mathbb{F}_2$ (the constraints are linear functions of the separator variables).

Step 5: Information gap

We have $\Omega(nq)$ independent constraints crossing the separator, but only $O(k)$ constraints resolvable from separator assignments.

The gap $\Omega(nq) - O(k) = \Omega(nq)$ (for $k=O(1)$) represents information that **must be communicated** in the message $M_{\{u \rightarrow v\}}$.

Since each independent constraint is one bit, the message requires $\Omega(nq)$ bits.

With k growing: More generally, the message requires $\Omega(nq/(k+1))$ bits, as the separator can resolve $O(k)$ constraints, leaving $\Omega(nq - k) \approx \Omega(nq/k)$ for $k = o(n)$.

This completes Lemma 3. ■

4.4 Consequence for Computation

Any algorithm using width- k decomposition must:

- Construct messages of size $\Omega(nq/(k+1))$ bits
- Process these messages (read/write)
- Total work $\geq \Omega(nq/(k+1))$ operations

For $k=O(1)$ and $q=\Theta(n)$: work $\geq \Omega(n^2)$.

For input size $N = \Theta(nq) = \Theta(n^2)$: work $\geq \Omega(N)$.

This is polynomial, but crucially: **the bound applies per instance** and cannot be reduced by clever algorithms while maintaining $k=O(1)$.

5. Integrated Result and Variants

5.1 Theorem A (Main Result)

Theorem A (SAT $\notin$ Rail-Admissible-Polytime).

No rail-admissible polytime evaluator with $k=O(1)$ decides SAT on the family $\{\phi_{\{G,n\}}\}$.

Proof:

Assume such an evaluator exists.

By Lemma 1: the computation has $tw = O(1)$.

By Lemma 2: the family $\{F_n\} \subseteq \{\phi_{\lfloor G, n \rfloor}\}$ has $tw = \Omega(n)$.

Contradiction: Cannot represent $\Omega(n)$ -treewidth graphs with $O(1)$ -treewidth computation.

Alternatively, by Lemma 3: even if we allow width k , communication requires $\Omega(n^2/k)$ bits. For $k=O(1)$, this is $\Omega(n^2)$ bits per heavy separator. With $\Theta(n)$ such separators (by tree structure), total communication = $\Omega(n^3)$ bits for input size $N=\Theta(n^2)$, implying $\Omega(n^3/N) = \Omega(n)$ operations per input bit, exceeding linear-time bounds.

Under either argument: **SAT $\notin$ (Polytime n Bounded-Separator).** ■

5.2 Variant B1: Bandwidth-Limited Model

Setting: Each separator can exchange at most $b=\text{poly}(k)$ bits per round; total rounds $\leq \text{polylog}(N)$.

Strengthened result: For $k=O(1)$, $b=O(1)$, Lemma 3 requires $\Omega(n^2)$ total bits. This exceeds $\text{polylog}(N) \cdot b = \text{polylog}(n^2) \cdot O(1) = O(\log^c n)$ for any c .

Conclusion: $\Omega(n^2/\log^c n)$ rounds required $\rightarrow$ super-polylog $\rightarrow$ **super-polynomial time**.

5.3 Variant B2: Aggregate Communication Budget

Setting: Total communication across all separators $\leq \tilde{O}(N)$ (near-linear).

Analysis: Centroid decomposition of tree T yields $\Omega(n)$ heavy edges each requiring $\Omega(n^2/k)$ communication by Lemma 3. For $k=O(1)$: total $\geq \Omega(n) \cdot \Omega(n^2) = \Omega(n^3) = \Omega(N^{\{3/2\}})$.

Conclusion: Exceeds near-linear budget $\rightarrow$ rules out near-linear-time evaluation.

5.4 Parameterized Tradeoff

Theorem B: For width $k(N)$, communication on $\phi_{\lfloor G, n \rfloor}$ requires $\Omega(N/k)$ bits at some separator.

Implications:

- $k = \text{polylog}(N)$: communication = $\Omega(N/\text{polylog}(N)) = \Omega(N) \rightarrow$ near-linear work
- $k = n^\alpha$ for $\alpha < 1$: communication = $\Omega(n^{\{2-\alpha\}}) \rightarrow$ super-linear in n
- $k = n$: communication = $O(n) \rightarrow$ linear (but this violates rail constraints)

This establishes a smooth tradeoff: reducing k (tighter geometric constraints) increases required communication.

6. Discussion of Model Equivalence

6.1 The Central Question

Does $P = k$ -Rail-Admissible for some $k = \text{polylog}(n)$?

This determines whether Theorem A implies $P \neq NP$.

6.2 Arguments for $P \not\subseteq \text{Rail-Admissible}$

Observation 1 (Space usage): Standard Turing machines in P can use polynomial space. This corresponds to separator size $O(n^c)$. Rail-admissible requires $k = \text{polylog}(n)$. Gap: polynomial vs polylog.

Example: Simple SAT algorithm uses $O(n)$ workspace (storing current assignment), taking exponential time. But a hypothetical P algorithm for SAT could use $O(n^2)$ space and polynomial time. This would have separator size $O(n^2)$, exceeding polylog bounds.

Observation 2 (Known classes):

- **NC** (Nick's Class): polylog-depth parallel circuits with polynomial processors. Has bounded separator size (depth = separator bound). $NC \subseteq P$, likely $NC \subsetneq P$.
- **SC** (Steve's Class): polylog-space sequential computation. $SC \subseteq P$, likely $SC \subsetneq P$.

Rail-Admissible appears similar to NC or SC, suggesting it's a proper subset of P .

Observation 3 (Circuit width): Logspace-uniform circuits (which equal P) can have width polynomial in n (number of wires between consecutive layers). Rail-admissible requires width $\text{polylog}(n)$. Again: polynomial vs polylog gap.

6.3 Potential Path to Model Equivalence

Conjecture (likely false): Every polynomial-time algorithm can be converted to polylog-separator form with at most polynomial overhead.

What this would require:

- Show any P algorithm has a "narrow" structure (bounded information flow at any time-step)
- Or show efficient simulation: polynomial-space $\rightarrow$ polylog-space with polynomial time overhead
- This would imply $P \subseteq SC$ (polylog-space), which is widely believed false

Barrier: Standard diagonalization suggests P -complete problems exist that inherently require polynomial space, preventing compression to polylog-separators.

6.4 Current Assessment

Most likely: $\text{Rail-Admissible} \subsetneq P$

Therefore: Theorem A proves $\text{SAT} \notin \text{Rail-Admissible}$, an interesting separation, but not $P \neq NP$.

Value: The result still identifies a natural geometric constraint (bounded treewidth) that SAT violates, providing new perspective on computational hardness.

7. Open Problems and Future Directions

7.1 Resolve Model Equivalence

Problem: Characterize exactly which complexity class equals $k(n)$ -Rail-Admissible for various $k(n)$.

Approaches:

- Compare to NC, SC, bounded-treewidth circuit classes
- Prove simulation theorems or separation results
- Identify P-complete problems provably outside Rail-Admissible

7.2 Extend to Other Problems

Problem: Do other NP-complete problems require unbounded treewidth?

Expectation: Yes, via standard reductions. Expander structure seems generic to NP-hardness.

Interesting cases:

- Problems with known bounded-treewidth instances (how do hardness patterns emerge?)
- #P problems (counting versions may have different treewidth requirements)

7.3 Algorithmic Applications

Problem: Can treewidth bounds guide SAT solver design?

Observation: Modern CDCL solvers implicitly manage clause width (related to treewidth). Our bounds suggest fundamental limits to "thin" reasoning strategies.

Application: Use treewidth to predict instance hardness; route search strategies to avoid high-treewidth regions.

7.4 Quantum Analogue

Problem: Define quantum rail-admissible computation (bounded entanglement across separators). Does BQP separate from quantum-rail-admissible?

Motivation: Quantum circuits have entanglement structure. Bounded entanglement = bounded separator in some sense. Could yield insights into BQP vs P.

7.5 Connection to Other Frameworks

The rail-admissible model emerged from a broader framework ("Geometric Grammaton Physics") attempting to unify:

- Complexity theory (dimensional stratification)
- Graph theory (treewidth as geometric constraint)
- Information theory (communication bounds)
- Optimization theory (adversarial robustness)

Open question: Can this framework be formalized to yield additional complexity-theoretic results? Does the "pentad structure" (D=5 optimality for adversarial robustness) have rigorous mathematical formulation with provable consequences?

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